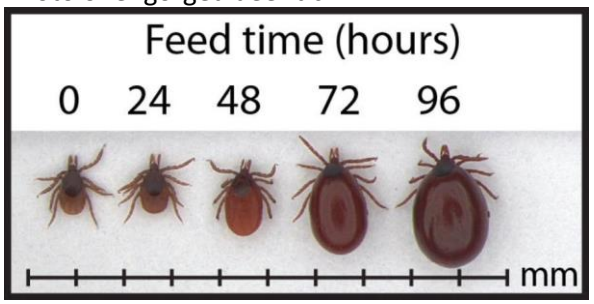
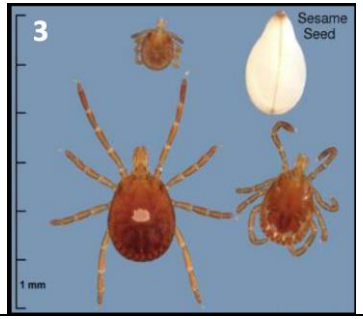
MICHIGAN MEDICINE
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Outpatient Treatment of Lyme Disease for Pediatric Patients

Type of Tick	Photo of Tick Stages	Diseases Tick May Carry	Comments
American Dog Tick *Dermacentor variabilis*		Lyme Disease None Rocky Mountain Spotted Fever, Tularemia Rare in Michigan, but possible	If you are in doubt of the type of tick, you can submit a Tick Pic to MDHHS for quick help in identifying the species. Or, if you have the actual tick, it can be submitted to the micro lab at Michigan Medicine for identification. Photo of engorged deer tick 
Blacklegged (Deer) Tick *Ixodes scapularis*		Lyme Disease Varies by county, now considered endemic Anaplasmosis, Ehrlichiosis, Babesiosis, Deer-tick virus Rare in Michigan, but possible	
Lone Star Tick *Amblyomma americanum*		Lyme Disease None Rocky Mountain Spotted Fever, tularemia, Ehrlichiosis, alpha-gal syndrome Rare in Michigan, but possible	

Patient Population	Empiric Therapy	Comments
High-risk tick exposure (all 3 criteria): 1. Bite was from an *Ixodes spp.* vector species (or unknown spp.) 2. It occurred in a highly endemic area (State of Michigan included) 3. Tick was attached for ≥ 36 hours	<u>Low-risk or equivocal-risk tick bite:</u> No antibiotics recommended. Watch and wait approach. <u>High-risk tick bite:</u> **Doxycycline** 4.4 mg/kg PO once (max: 200 mg/dose) To be given within 72 hours of tick exposure Amoxicillin prophylaxis has not been studied sufficiently but likely would require a longer course than doxycycline because of its shorter half-life and is not recommended.[2]	<u>Diagnosis and Testing</u> • Decision to treat EM is based on presence of a characteristic lesion and exposure in an endemic area. Serologies are often negative with a single EM lesion. • In the setting of atypical lesions, recommend acute phase Lyme serology, and if negative repeat convalescent testing (2-3 weeks later). • Skin lesions resolve within several days of therapy; if no improvement, consider alternative diagnoses. <u>Antimicrobial Selection and Dosing</u> • Doxycycline can be used for short durations (21 days or less) without regard to patient age.[2] • Provide patient education for doxycycline: avoid sun exposure, do not take with dairy or divalent/trivalent cations (multivitamins, supplements), do not lay down 30 minutes after administration of tablets. • Cefuroxime suspension is not commercially available. • See β-lactam Allergy Evaluation and Empiric Therapy Guideline for patients with β-lactam allergies. • Adjust amoxicillin and cefuroxime for renal dysfunction. <u>Other treatment considerations</u> • Consider co-infection with Babesia or Anaplasmosis if leukopenia, neutropenia, anemia, thrombocytopenia, elevated Tbili, elevated LDH, fever persisting > 1 days from start of antibiotics. Coinfection with Anaplasma can be treated with doxycycline. If fever persists despite treatment with doxycycline, B. microti infection is an important consideration, which requires alternative treatment. • If symptoms and signs of cardiac involvement (dyspnea, edema, palpitations, lightheadedness, chest pain, or syncope), obtain screening ECG. If abnormal from baseline consistent with lyme carditis, consider hospitalization. Per IDSA Guidelines, "in patients with or at risk for severe cardiac complications of Lyme disease including those with significant PR prolongation (PR > 300 milliseconds), other arrhythmias, or clinical manifestations of myopericarditis, we recommend hospital admission with continuous ECG monitoring". • For patients who have persistent or recurring nonspecific symptoms following recommended treatment for Lyme disease, but who lack objective evidence of reinfection or treatment failure, additional antibiotic therapy is not recommended.
Erythema migrans (EM) (single or multiple lesions)	<u>Preferred:</u> **Doxycycline** 2.2 mg/kg/dose PO BID (max: 100 mg/dose) x10 days <u>Allergy or intolerance to doxycycline:</u> **Amoxicillin** 16.67 mg/kg/dose PO TID (max: 500 mg/dose) x14 days <u>Allergy to doxycycline AND low/medium-risk PCN allergy:</u> **Cefuroxime** 15 mg/kg/dose PO BID (max: 500 mg/dose) x14 days <u>Allergy to doxycycline AND high-risk PCN AND cephalosporin allergy:</u> **Azithromycin** 10 mg/kg/dose PO once daily x7 days (max: 500 mg/dose)	

Lyme arthritis	An oral agent as for EM (except azithromycin), for 28 days	• With persistent arthritis after course of therapy, recommend Pediatric Infectious Diseases involvement if not already done. Can consider repeat course of oral therapy versus ceftriaxone for 14-28 days. • Approximately 10% to 15% of patients treated for Lyme arthritis will develop persistent synovitis that can last for months to years; referral to a rheumatologist may be appropriate for more severe cases.
Isolated facial palsy	**Doxycycline** 2.2 mg/kg/dose PO BID (max: 100 mg/dose) x14–21 days	• Evaluate for other possible causes (such as HSV); Lyme serology is typically positive at this stage. • Per IDSA Guidelines: As the diagnosis of Lyme neuroborreliosis may not be obvious at the time of presentation with a facial nerve palsy and because corticosteroids are most effective in idiopathic facial nerve palsy if administered within the first 72 hours after onset, corticosteroids should be instituted immediately in patients in whom the diagnosis of Lyme disease is uncertain. When the diagnosis of Lyme disease becomes apparent, the decision to stop corticosteroids that have already been started, or to start them in a patient initially presenting with acute Lyme disease-associated facial palsy, is a matter of patient preference and clinical judgment. • Doxycycline is preferred regardless of age; amoxicillin is unlikely to reach therapeutic levels in the central nervous system.
Other Lyme manifestations (such as carditis, meningitis)	Consult Pediatric Infectious Diseases	

Last revised: 08/2024	**Antimicrobial Stewardship Subcommittee:** 09/2024	**P&T:** 10/2024

References

	Citation
1	Lantos et al. Clinical Practice Guidelines by the Infectious Diseases Society of America (IDSA), American Academy of Neurology (AAN), and American College of Rheumatology (ACR): 2020 Guidelines for the Prevention, Diagnosis and Treatment of Lyme Disease. Clin Infect Dis. 2021 Jan 23;72(1):e1-e48.
2	Red Book: 2024–2027 Report of the Committee on Infectious Diseases (33rd Edition). By. Committee on Infectious Diseases, American Academy of Pediatrics.

Revision History

Date	Revision

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time-to-time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.